

Science & Biology Sample Document

1. Cellular Biology and Bioenergetics: The cell is the fundamental unit of structure and function in living organisms. Prokaryotes lack membrane-bound nuclei, whereas eukaryotes contain distinct organelles such as mitochondria, which synthesize adenosine triphosphate (ATP) through cellular respiration. Ribosomes translate genetic code into proteins.
2. Photosynthesis & Molecular Genetics: Photosynthesis occurs inside plant chloroplasts where light energy converts water and carbon dioxide into glucose and oxygen gas. DNA (deoxyribonucleic acid) utilizes adenine, thymine, cytosine, and guanine nitrogenous base pairs to store genetic data in a double helix structure.
3. Ecosystem Ecology & Human Systems: Trophic structures direct energy flow from primary autotrophic producers to secondary consumers. Decomposers recycle essential organic nitrogen and carbon. In humans, deoxygenated blood returns to the right heart atrium before entering lungs via pulmonary arteries.